

SAP CPI Project Documentation

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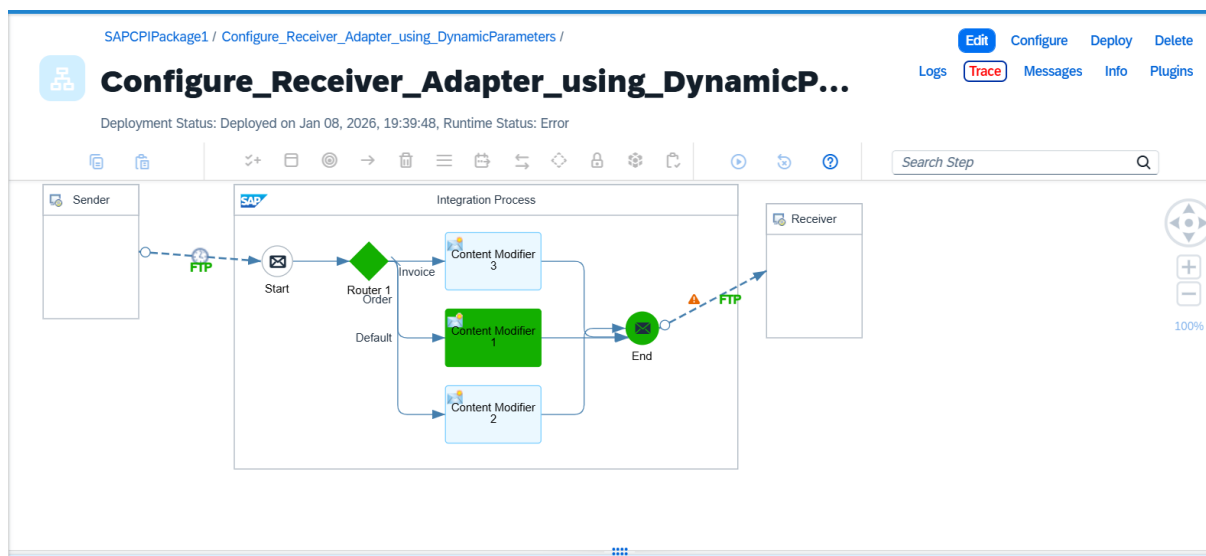
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Scenario: Content-Based File Passthrough Interface

1.Scenario Overview

This project automates the classification and routing of inbound files received from an upstream system. Files are transferred to SAP Cloud Platform Integration (CPI) via FTP. Based on the file name pattern, SAP CPI intelligently routes the files to different target folders on the FTP server. Files starting with “Order” are routed to the Orders folder, files starting with “Invoice” are routed to the Invoices folder, and all other files are moved to a default Other folder. This solution eliminates manual file segregation and ensures organized downstream processing.

2.Architechture Diagram



3. Step-by-Step Implementation

- Configured FTP Sender Adapter to receive files from the source system
- Implemented Content-Based Routing (CBR) using Regex Expressions
- Dynamically determined the target directory based on business rules:
 - Files starting with “Order” → Orders folder
 - Files starting with “Invoice” → Invoices folder
 - All other files → Other folder
- Configured FTP Receiver Adapter with dynamic directory assignment using message properties.

- Performed end-to-end testing for multiple file name scenarios to ensure accurate routing.

Step1 – Configure the FTP Sender Adapter:

Fetching all files in directory “ALL_FILES”

FTP Adapter (Sender):

FTP			
General	Source	Processing	Scheduler
FILE ACCESS PARAMETERS			
Directory:	ALL_FILES		
File Name:	*		
CONNECTION PARAMETERS			
Address:	ftp.drivehq.com		
Proxy Type:	Internet		
Encryption:	Explicit FTPS		
Credential Name:	FTP_Connection		
Timeout (in ms):	10000		
Maximum Reconnect Attempts:	3		
Reconnect Delay (in ms):	30000		

Step2 – Configure Content -Based Router

The filename of the incoming file will be available in the header parameter CamelFileNameOnly”.

We will route the files based on the prefix of the filename. Using a Regex expression, We can find if the filename matches the pattern we are looking for

Regular expression to check if the file name starts with Order. Or Invoice.

`${header.CamelFileNameOnly} regex 'Order.*'`

`${header.CamelFileNameOnly} regex 'Invoice.*'`

Router:

Router

General Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Order	<code>\${header.CamelFileNameOnly} regex 'Order.*'</code>	☐
2	Default		☒
3	Invoice	<code>\${header.CamelFileNameOnly} regex 'Invoice.*'</code>	☐

Step3 – Make Use of Exchange or Header parameters to set the Directory

Let's make use of content modifiers to determine the directory at runtime.

We will have an exchange property parameter named "directory" to set the value of the directory.

Content Modifier:

Content Modifier

General Message Header Exchange Property Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	directory	Constant	Order		

Content Modifier

General Message Header Exchange Property Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	directory	Constant	Invoice		

Step4 – Configure Receiver Adapter using Dynamic Parameters

We will make use of Camel Expressions to define the filename dynamically.

`${file:onlyname.noext}` – Camel Simple Expression to get the prefix or the filename of the incoming file without the extension

FTP Adapter (Receiver):

FTP

General

Target

Processing

FILE ACCESS PARAMETERS

Directory:

Property

Value

Directory

File Name:

Property

Value

File Name

Append Timestamp:

Property

Value

☒

CONNECTION PARAMETERS

Address:

Property

Value

Address

Proxy Type:

Property

Value

Internet

Encryption:

Property

Value

Explicit FTPS

Credential Name:

Property

Value

Credential Name

Timeout (in ms):

Property

Value

Timeout

Maximum Reconnect Attempts:

Property

Value

Maximum Reconnect Attempts

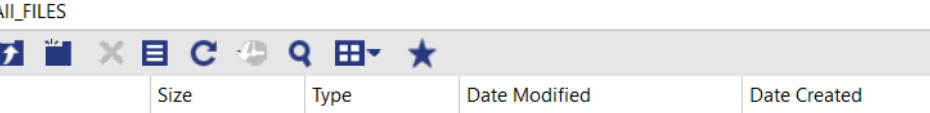
Reconnect Delay (in ms):

Property

Value

Reconnect Delay

6. Input Data:



Name ▲	Size	Type	Date Modified	Date Created
InvoiceA.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
InvoiceB.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
InvoiceC.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
Order details1.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
Order details2.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
Order details3.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
POOrders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
PO_Orders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50
Sales_Orders.txt	13 B	Text Document	2026-01-08 19:01:31	2026-01-10 19:06:50

The screenshot shows the 'My Storage (\)' window with the following data:

Name	Size	Type	Date Modified	Date Created
UniqueFiles	578 B	File	2026-01-10 15:59:23	2026-01-10 15:59:23
DriveHQData	3 KB	Online Backup ...	2026-01-10 12:16:16	2026-01-10 12:16:15
Express_Orders	1023 B	File Folder	2026-01-10 12:17:27	2026-01-10 12:17:26
FoodDelivery	662 B	File Folder	2026-01-10 14:00:16	2026-01-10 13:57:41
FoodDelivery_1	662 B	File Folder	2026-01-10 14:00:16	2026-01-10 13:57:41
Instamart	311 B	File Folder	2026-01-10 14:04:20	2026-01-10 14:04:19
Instamart_1	311 B	File Folder	2026-01-10 14:04:20	2026-01-10 14:04:19
Invoice	26 B	File Folder	2026-01-08 19:34:42	2026-01-08 19:34:32
My Documents	138 B	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45
My Pictures	3,021 KB	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45
Online_Orders	1022 B	File Folder	2026-01-10 12:15:26	2026-01-10 12:15:24
Order	39 B	File Folder	2026-01-08 19:39:57	2026-01-08 19:39:53
Others	52 B	File Folder	2026-01-08 19:40:02	2026-01-08 19:34:22
Recycle Bin	3 KB	File Folder	2026-01-10 12:20:17	2026-01-10 12:20:12
UniqueFiles1	578 B	File Folder	2026-01-10 16:19:42	2026-01-10 16:19:41
wwwhome	24 KB	File Folder	2026-01-08 19:30:45	2026-01-08 19:30:45

A screenshot of a Windows File Explorer window. The address bar shows the path '\Order'. The ribbon at the top includes icons for Back, Forward, Up, Down, Home, and Favorites. The main area displays a table of files and folders. The table has five columns: Name, Size, Type, Date Modified, and Date Created. There are three items listed: 'Order details2026010814...', 'OrderC20260108140954', and 'OrdersA20260108140956'. Each item is represented by a document icon.

A screenshot of a Windows File Explorer window. The address bar shows the path '\Invoice'. The ribbon menu is visible with icons for Back, Forward, Up, Down, Delete, Share, Refresh, View, Search, and Favorites. The main area displays a table of files with columns: Name, Size, Type, Date Modified, and Date Created. Two files are listed: 'InvoiceB20260108140431' and 'InvoiceC20260108140441', both of size 13 B, type File, and modified on 2026-01-08 at 19:34:32 and 19:34:41 respectively.

 \Others					
          					
Name ▲	Size	Type	Date Modified	Date Created	
 InvoiceA20260108140421	13 B	File	2026-01-08 19:34:23	2026-01-08 19:34:23	
 POOrders20260108141000	13 B	File	2026-01-08 19:40:00	2026-01-08 19:40:01	
 PO_Orders202601081409...	13 B	File	2026-01-08 19:39:58	2026-01-08 19:39:59	
 Sales_Orders2026010814...	13 B	File	2026-01-08 19:40:02	2026-01-08 19:40:02	

8. Iflow Package:



Configure_Receiver_Adapter_using_DynamicParameters.zip